

Written Status Report

CMSI 4072 Project Status Sheet Spring 2026

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Project: Sammy's Games: A Board and Card Game Platform

Date: March 25, 2026

Period Covered: March 11 – March 25

Accomplishments since last report:

Completed the core mechanical requirements for **Six Hands**, transforming it from a visual prototype into a fully functional logic puzzle. Specific technical milestones include:

- **Connectivity Validation:** Successfully implemented a Breadth-First Search (BFS) algorithm (isConnected) to enforce the "orthogonal connectedness" rule. The validator now correctly identifies if a color group is fragmented.
- **Poker Hand Evaluation:** Integrated a robust hand-ranking engine that handles standard poker hands and accounts for **Jokers** as wild cards by iterating through the deck to find the optimal substitution.
- **Dynamic Scoring & Feedback:** Implemented a "Lowest Hand" identification system. The UI now highlights the weakest of the six hands in gold, as the player's primary goal is to maximize the strength of their "weakest link."
- **UI/UX Refinement:** * Enhanced the **Color Picker** with a dedicated "Sixth Hand" (Star) toggle.
 - Added dynamic hand-name labeling (e.g., "Full House", "Straight") in the side panel that updates in real-time.
 - Improved card visuals with distinct CSS classes for suits (Red, Blue, Green, Black) and a unique gradient for Jokers to ensure the grid is scannable at a glance.
- **Game State Logic:** Finalized the daily seed generation logic, ensuring that the 6x5 grid is consistent for all players on a given calendar day.

Scheduled tasks to be done by next report:

- **Global Leaderboards:** Implement a backend integration to allow players to submit their "Lowest Hand" score and compare it against the daily global average (April 8).
- **Mobile Optimization:** Refine CSS Grid layouts and touch-targets for better playability on mobile browsers, as the current 6x5 grid is slightly too wide for standard portrait views (April 8).

- **Game #3 Scoping:** Finalize the concept for the third and final game. Current front-runners are a tile-matching game or a simplified trick-taking game (April 8).

Noteworthy risks, concerns, or problems:

- **Risk: Complexity of Game #3:** With the semester progressing, there is a risk that a complex third game will detract from polishing Six Hands and Knockout.
 - Mitigation Plan: I will prioritize a "high-polish, low-complexity" mechanic for the final game to ensure the entire platform feels cohesive and bug-free.
- **Concern: Performance on Old Devices:** The BFS connectivity check and Joker evaluation (which iterates the deck) run on every click. While smooth on desktop, it may cause lag on older mobile devices.
 - Mitigation Plan: If performance issues arise, I will move the validation logic to a `requestAnimationFrame` or debounce the function to ensure the UI remains responsive.

Current Status: ON TRACK

Six Hands is now feature-complete regarding its core rules. The project is shifting from "foundational coding" to "polish and platform integration."